

SARVESH DATAR

Chemical Process Engineer · Crystallisation & Separation Processes · Mechanistic-ML Integrated Modelling

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RESEARCH INTERESTS

Crystallisation and industrial separation processes (distillation, pressure-swing adsorption); integration of machine learning with mechanistic and population balance models for design and control of crystallisation and separation systems; equation-oriented modelling and optimisation-based process design; and process intensification (reactive and pressure-swing distillation).

EDUCATION

MSc Advanced Chemical Engineering with Process Systems Engineering

2025

Imperial College London — Result: Distinction

- Thesis: “Predicting and Controlling the Growth Shape of Crystals: A Solvent Selection Approach,” supervised by Prof. Claire Adjiman.
- Relevant coursework: Advanced Process Design, Advanced Process Optimisation, Dynamic Behaviour of Process Systems, Advanced Process Operations, Machine Learning for Chemical Engineering, Pharmaceutical Process Development, Sustainable Energy Technologies.

B.Tech, Chemical Engineering (Minor in Computer Science Engineering)

2024

MIT World Peace University, Pune — CGPA: 9.92 / 10, Gold Medalist

- Additional coursework: Artificial Intelligence and Machine Learning, Database Management Systems, Bioreactor Design and Analysis, Biocatalysis, Software Development and Project Management.

PUBLICATIONS

Narayan, G., Datar, S. et al. “Effect of Sulfonation Density on Acid Strength in Ion Exchange Resins: Insights from Solid-State NMR and Density Functional Theory,” *Molecular Catalysis*, Elsevier, 2024. DOI: 10.1016/j.mcat.2026.115794.

Datar, S. et al. “Experimental Study on Green Synthesis of Copper Nanoparticles and Its Utilisation for Degradation of Organic Dye,” *Springer Lecture Notes in Civil Engineering*, 2023. DOI: 10.1007/978-981-99-6762-9_7.

RESEARCH EXPERIENCE

Master's Thesis — Crystal Growth Shape Prediction and Control

2025

Department of Chemical Engineering, Imperial College London | Supervisor: Prof. Claire Adjiman

- Developed a mechanistic model linking solvent selection and operating conditions to crystal growth shape, and validated it against experimental data for naphthalene and biphenyl systems.
- Extended the model into a design tool for selecting solvents and conditions that avoid undesirable crystal morphologies, and built a complementary machine-learning model trained on experimental and literature data.

Research Intern — Active Pharmaceutical Ingredient Precursor Scale-Up

2024

Indian Institute of Technology, Bombay (IIT Bombay)

- Scaled a lab process to a pre-pilot facility (packed-bed reactor, batch distillation, in-line GC monitoring) and studied the effect of catalyst sulphonation density on acid activity and selectivity — the basis for the Molecular Catalysis publication.
- Cut production cost by 40% by introducing biorefinery-based raw materials and a reusable solid catalyst, alongside optimised process parameters; proposed a reactive-distillation route to replace the reactor-distillation train.

Project Trainee — Process Design for a Perfumery Intermediate

2023

CSIR-National Chemical Laboratory (CSIR-NCL)

- Investigated lab- and pilot-scale production of acetophenone via ethylbenzene and cumene hydroperoxide oxidation, identifying process-control requirements to intensify the route safely (PFR/CSTR and reactive distillation) given the instability of the hydroperoxide intermediates.

PROFESSIONAL EXPERIENCE

Associate Process Engineer, Equinox Software Services Pvt. Ltd.

2025 – Present

- Process design and optimisation for Jubilant Ingrevia: modelled 12 distillation columns (Aspen Plus and AspenBatchSep), then evaluated retrofit and green-field alternatives, including pressure-swing and reactive distillation, projecting a 60% cut in energy consumption and a 15% gain in product recovery.
- Contributed to R&D on a proprietary multi-batch distillation software package, studying time-marching and optimisation algorithms and prototyping AI/ML-based improvements — integrating machine-learning surrogates with mechanistic column models to accelerate convergence — validated against literature and legacy-software benchmarks.
- Built a lactic-acid fermentation model in Python, coupled it to BioSteam as an open-source alternative to proprietary simulators, and cross-validated it against an Aspen Plus model for multi-objective optimisation studies.
- Automation feasibility study for a 9-product, 3-line multi-batch facility, covering process flow, equipment, and control-system requirements, and their impact on product quality, safety, and regulatory compliance.

SELECTED INDEPENDENT & ACADEMIC PROJECTS

Equation-Oriented Distillation Column Simulator

2026

- Built a rigorous steady-state distillation simulator (Python, Pyomo, IPOPT) that solves the full MESH equations simultaneously via an equation-oriented NLP formulation; implemented physically informed initialisation, adaptive solver handling, and a custom bisection

- scheme for purity-constrained specifications to improve convergence.
- Open-source research platform (GitHub) intended for column optimisation, surrogate-model generation, and process-intensification studies.

Sustainable Formaldehyde Production from CO₂ and H₂

2024

Design Project, Imperial College London

- Designed a two-step catalytic route (CO₂/H₂ → methanol → formaldehyde) in Aspen HYSYS, applied heat integration to minimise utility demand, and conducted a life-cycle assessment in OpenLCA to quantify environmental impact.

Rapid Pressure-Swing Adsorption for Medical Oxygen Concentration

2023

Seminar Project, MIT-WPU

- Assessed the feasibility of a single-bed rapid PSA process as a lower-cost alternative to conventional two-bed VSA/PSA systems, optimising for oxygen recovery, energy use, and adsorbent cost; design was carried forward into a tested prototype.

AIChE ChemE Car — Thermoelectric-Generator-Powered Vehicle

2022

- Co-designed a car powered by exothermic/endothermic reactions driving a thermoelectric generator, with an iodine-clock reaction for stopping-distance control; served as team Safety Head. Won the Student Regional Conference 2022 and represented India at the AIChE Annual Student Conference.

Green Synthesis of Copper Nanoparticles for Dye Degradation

2022

- Developed a neem-leaf-extract-based, toxic-reagent-free synthesis route for copper nanoparticles (50–70 nm, confirmed by UV-Vis, FTIR, SEM), achieving ~60% photocatalytic degradation of methylene blue within 120 minutes.

TECHNICAL SKILLS

Process simulation & modelling: gPROMS, GAMS-style NLP/MINLP formulations, Aspen Plus, Aspen HYSYS, AspenBatchSep, BioSteam, Python (Pyomo, NumPy/SciPy), IPOPT, equation-oriented modelling

Programming & data: Python, SQL, machine learning fundamentals (scikit-learn), Git/GitHub

Other: OpenLCA (life-cycle assessment), MATLAB, Microsoft Office / process documentation and technical reporting

HONOURS & AWARDS

- Gold Medalist, B.Tech Chemical Engineering, MIT-WPU (2024) — CGPA 9.92/10
- Winner, AIChE Student Regional Conference ChemE Car Competition (2022); India representative, AIChE Annual Student Conference (2022)
- Distinction, MSc Advanced Chemical Engineering with Process Systems Engineering, Imperial College London (2025)